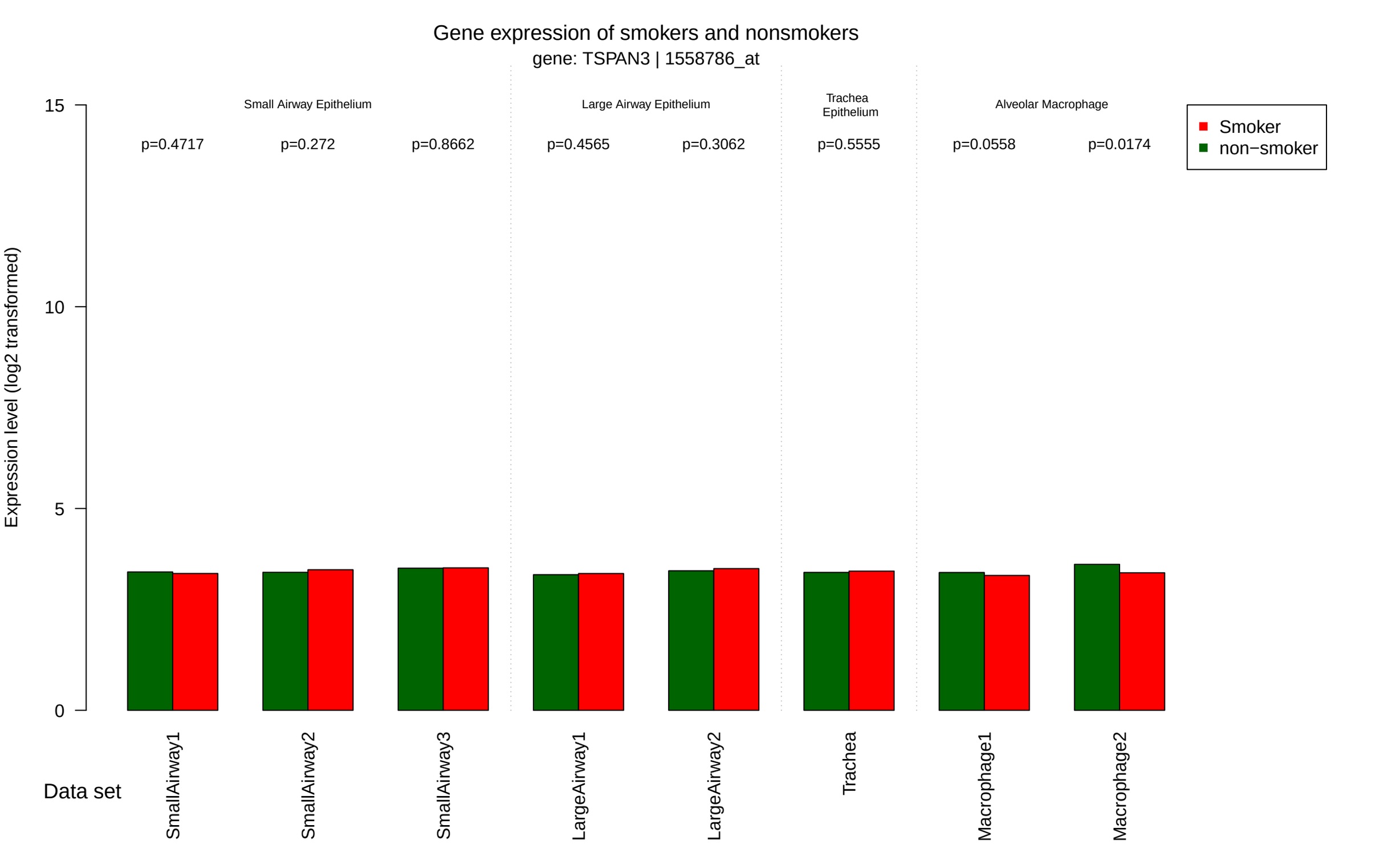
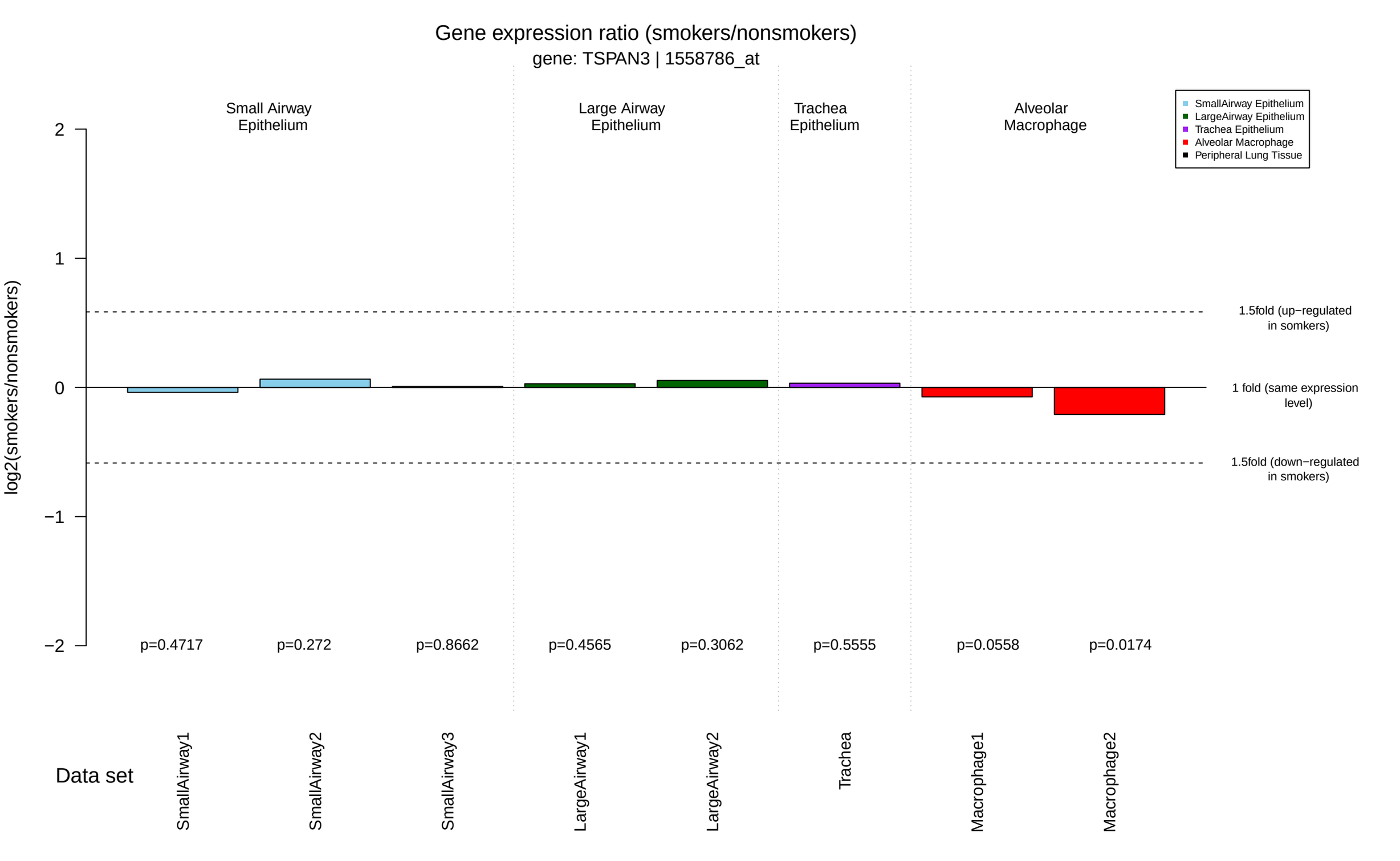
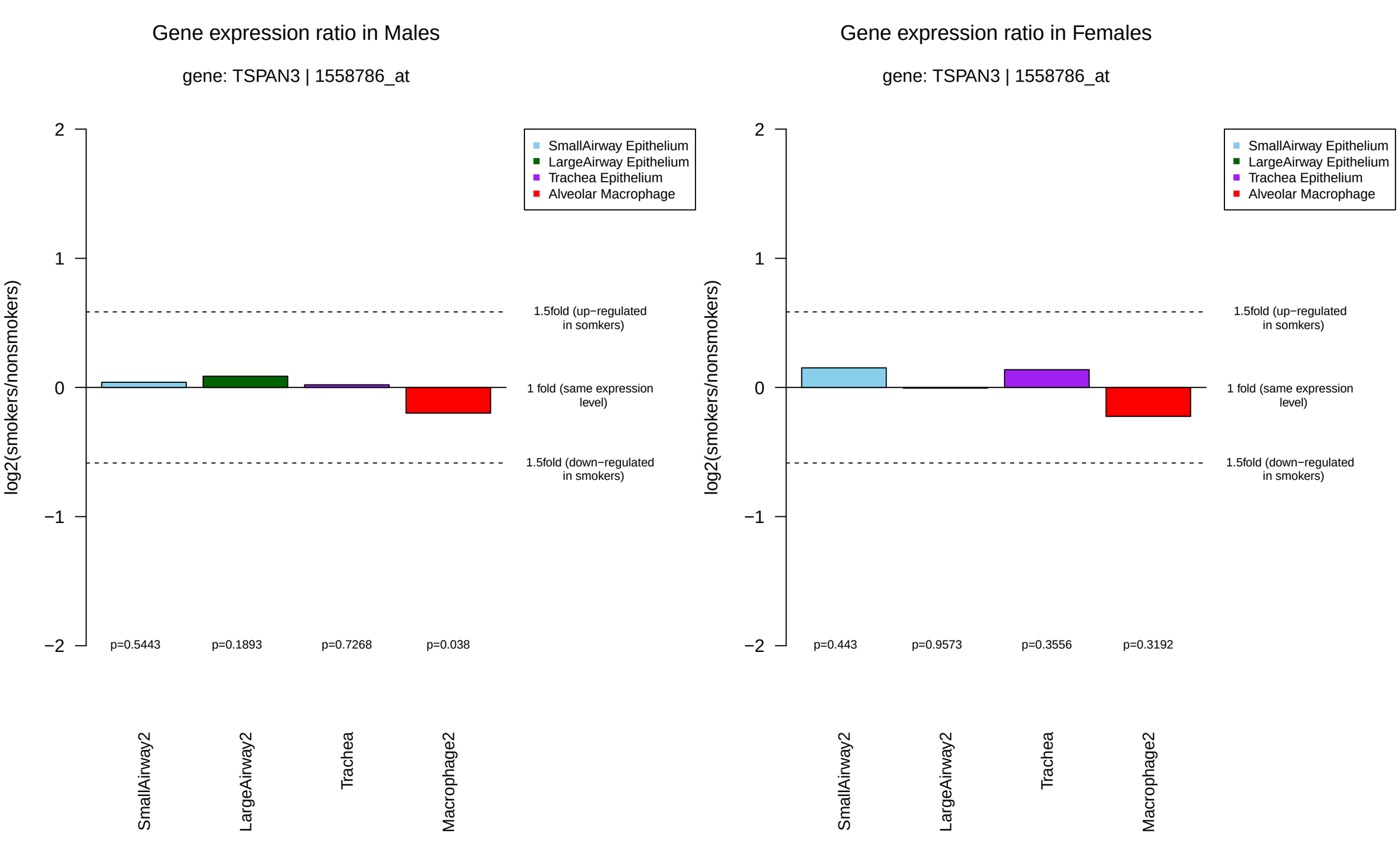
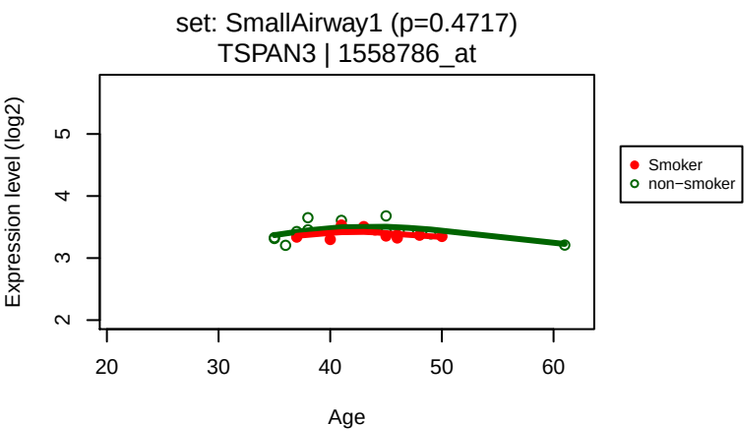
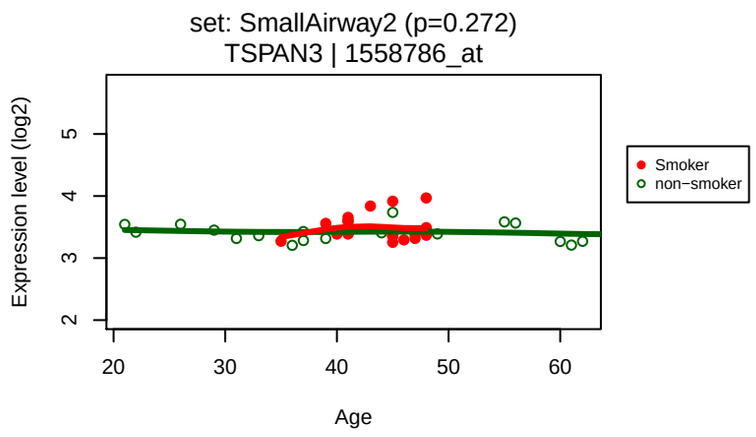
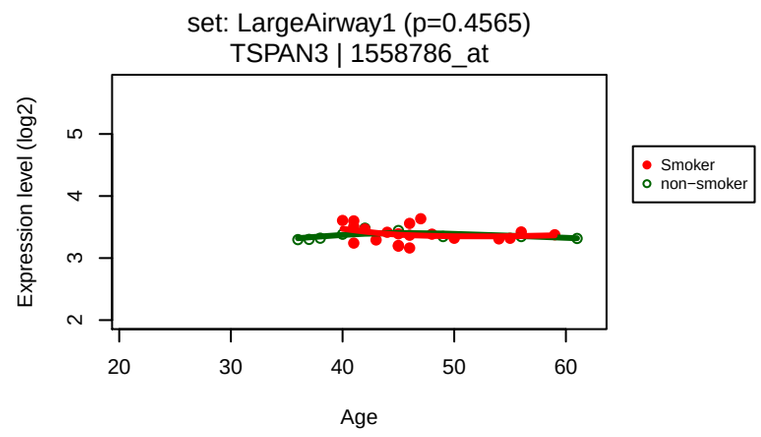

Gene expression of smokers and nonsmokers
gene: TSPAN3 | 1558786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4717
p=0.272
p=0.8662
p=0.4565
p=0.3062
p=0.5555
p=0.0558
p=0.0174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

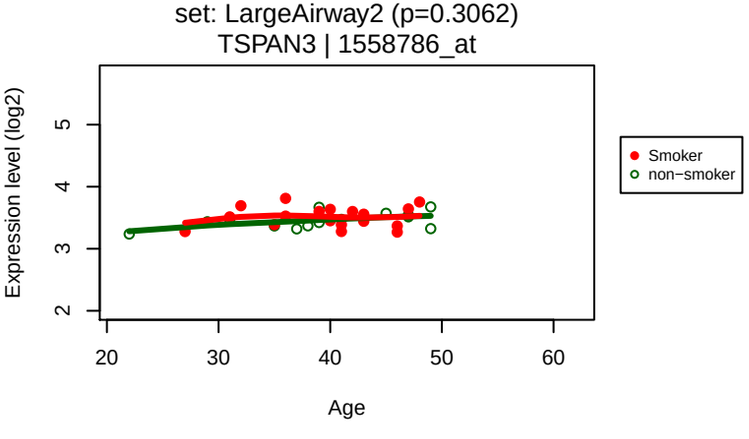

Gene expression ratio (smokers/nonsmokers)
gene: TSPAN3 | 1558786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4717
p=0.272
p=0.8662
p=0.4565
p=0.3062
p=0.5555
p=0.0558
p=0.0174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

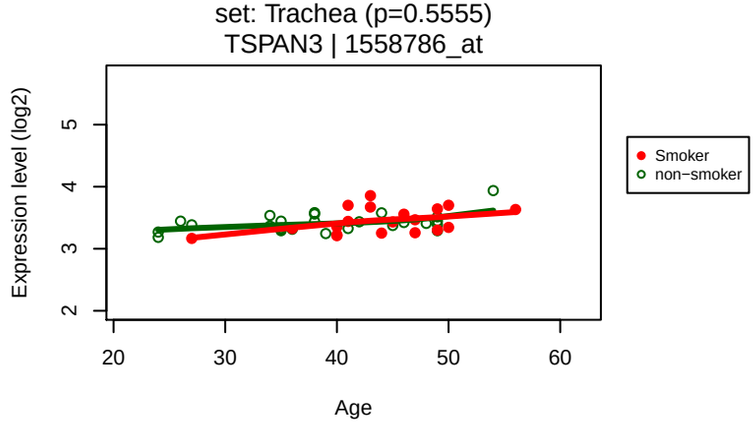

Gene expression ratio in Males
gene: TSPAN3 | 1558786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5443
p=0.1893
p=0.7268
p=0.038
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPAN3 | 1558786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.443
p=0.9573
p=0.3556
p=0.3192
SmallAirway2
LargeAirway2
Trachea
Macrophage2

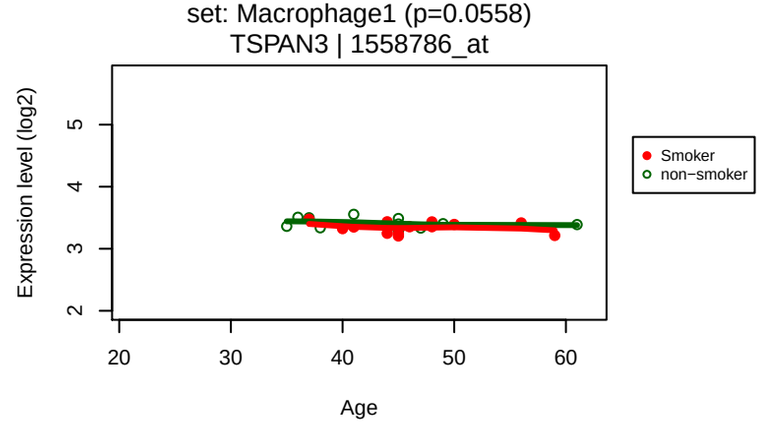

set: SmallAirway1 (p=0.4717)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker

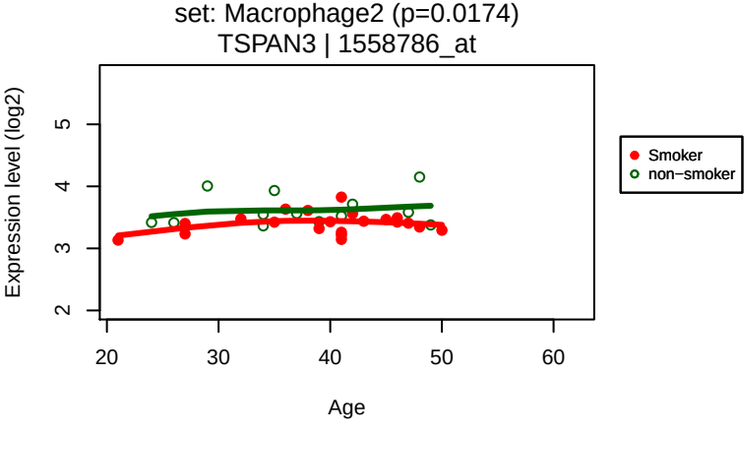

set: SmallAirway2 (p=0.272)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker

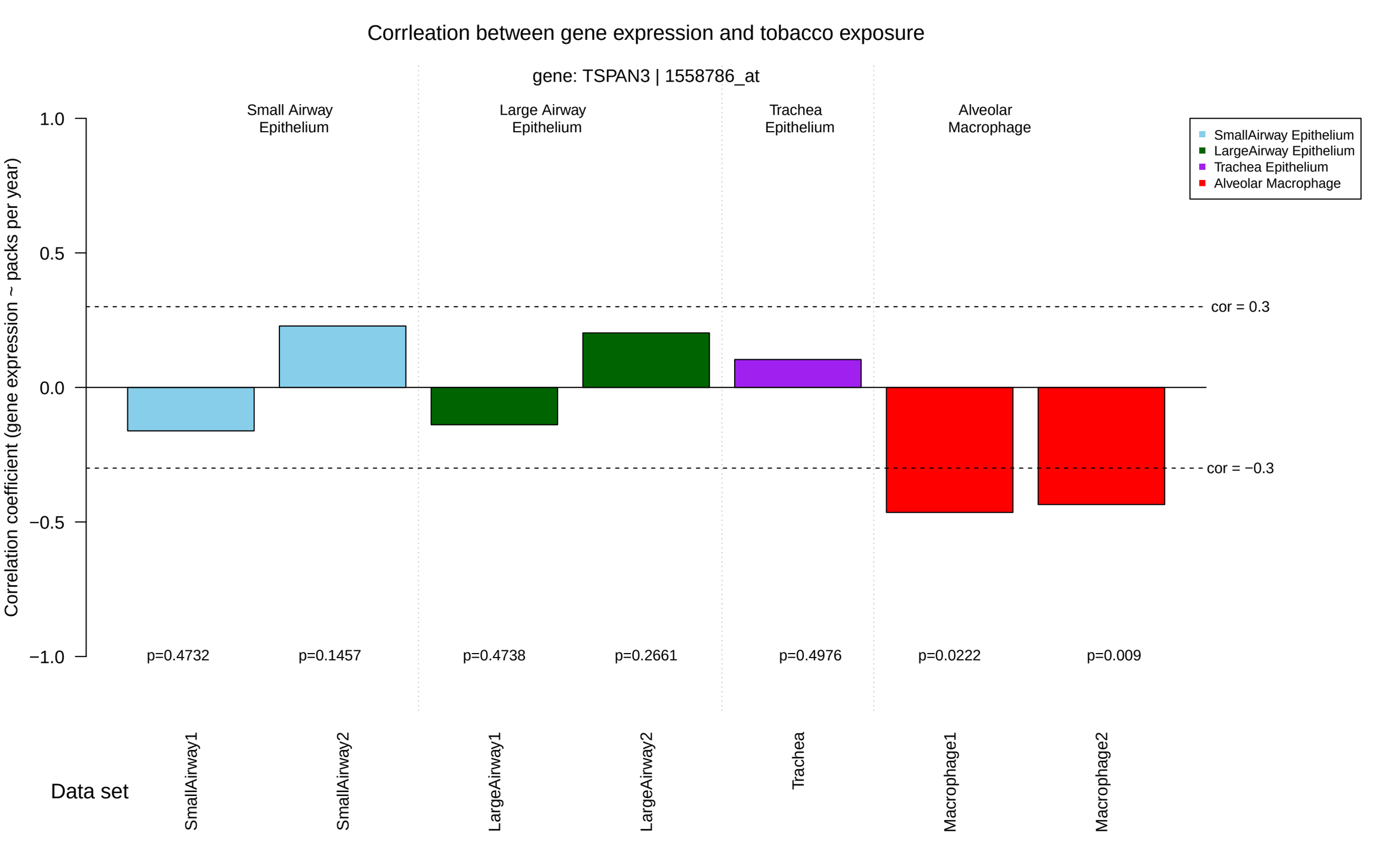

set: LargeAirway1 (p=0.4565)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3062)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5555)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0558)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0174)
TSPAN3 | 1558786_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPAN3 | 1558786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4732
p=0.1457
p=0.4738
p=0.2661
p=0.4976
p=0.0222
p=0.009
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2